

In-Fibre Optical Logic Gate Utilizing Laser-Inscribed Multi-Mode Interference within Multi-Core Fibre

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Abstract *We propose the in-fibre logic gate based on laser-inscribed multi-mode interference (MMI) within multi-core fibre to realize core switching by optical phase for the first time. The MMI-based logic gate is experimentally demonstrated, and we show its applicability to 2x4 optical logic circuits. ©2026 The Author(s)*

Introduction

Interest in optical computing and optical signal processing has been increasing from the viewpoints of low power consumption and high-speed computation [1–2]. Previously reported all-optical logic devices can be broadly classified into linear optical logic circuits, which are realized by optical interference and coupling, and nonlinear optical logic circuits, whose switching behaviour depends on input light intensity [3–4]. Linear optical logic circuits have attracted particular interest thanks to their low power consumption and ability to provide fixed logical mappings and specific signal-processing applications [3].

Linear optical logic circuits based on planar photonic waveguides are composed of combinations of functional elements capable of optical splitting/combining and phase shifting, enabling scalable circuit designs through parallelization and cascading [3–4]. Fibre-based linear optical logic circuits have also been studied due to advantages including low propagation loss and alignment-free operation [5–6]. However, in most fibre-based approaches, logic functions are defined by interference states formed within individual structures, making independent amplitude and phase control difficult and limiting scalable parallel or cascaded configurations. Multi-core fibres (MCFs) [7] provide multiple spatially separated optical paths within a single cladding, making them a highly integrated platform for optical circuits. Meanwhile, femtosecond-laser inscription enables localized functions to be written into selected cores of the MCF. This combination is well suited to scalable in-fibre optical circuits, in which different functions can be assigned to individual cores and cascaded along the fibre.

In this study, we combine an MCF platform with femtosecond-laser inscription to create a 2×2 linear optical logic gate. Femtosecond-laser-induced refractive-index modification allows functional regions to be written inside the fibre without cutting or splicing, thus enabling three-dimensional integration of linear optical logic functions.

We demonstrate 2×2 operation, featuring

amplitude weighting under single-input excitation and $\Delta\phi$ -controlled output-core switching under dual-input excitation. We also verify the scalability of the approach by extending the basic gate to a 2×4 in-fibre linear logic circuit.

Concept of MCF-integrated linear logic circuit using MMI-assisted coupler

In general, linear optical logic circuits define input–output mappings by exploiting optical splitting, coupling, and interference. Figure 1(a) shows the proposed in-fibre linear logic circuit implemented in an MCF, integrating four 2×2 logic gates and a single in-fibre phase-control section. In this circuit, the output cores (E–H) provide multiple logic outputs, with E corresponding to OR, F to XOR, and G and H to AND.

Two input signals, A and B, are launched into separate cores and processed by a sequence of 2×2 logic gates formed inside the fibre. The first logic gate provides asymmetric amplitude weighting of the two input signals, defining the relative contributions of A and B to subsequent stages. The downstream logic gates operate as balanced interferometric combiners in which the two weighted signal paths interfere with a controlled relative phase. As a result, different interference conditions are established at each output core, enabling the four outputs (E–H) to exhibit distinct logical behaviours determined by constructive or destructive interference.

Cascading such localized 2×2 logic gates enables multi-output linear optical logic circuits. For stable cascaded operation, each 2×2 logic gate must satisfy two essential requirements: (i) amplitude weighting under single-input excitation, and (ii) phase-dependent output switching under simultaneous dual-input excitation [3].

Figure 1(b) shows the structure of the proposed in-fibre logic gate, which consists of an MMI-assisted coupler and an in-fibre phase-control section. The MMI-assisted coupler forms a localized multimode interference (MMI) region between two single-mode cores, enabling controlled coupling and interference within a limited

fibre length. Under single-core excitation, the MMI-assisted coupler functions as a splitter that provides amplitude weighting. Under dual-core excitation, the output distribution depends on the relative phase difference $\Delta\phi$ between the two optical fields. When $\Delta\phi = 0$, the optical power is distributed nearly equally between the two outputs, whereas varying $\Delta\phi$ changes the interference condition and induces output-core switching. The inter-core phase difference $\Delta\phi$ is controlled by a phase shifter formed in one of the cores utilizing femtosecond-laser-induced refractive index modulation, enabling arbitrary phase shifts from 0 to 2π [8].

Figures 2 and 3 show the BPM simulation results at a wavelength of 1550 nm for the 1×2 and 2×2 operations, respectively. As shown in Fig. 2, under single-core excitation, the field evolution exhibited balanced splitting as designed. As shown in Fig. 3, under dual-core excitation, the output was distributed approximately 50:50 at $\Delta\phi = 0$ (blue line), whereas at $\Delta\phi = \pi/2$, the output was predominantly directed to Core 1. In addition, although not shown in the figure, the output was predominantly directed to Core 2 at $\Delta\phi = \pi$, confirming phase-dependent output switching. For a representative design ($W = 22 \mu\text{m}$, $L = 9400 \mu\text{m}$), the calculated excess loss was 0.026 dB, and the deviation from the target 50:50 splitting ratio was within $\pm 2.4\%$ (x-pol.) and $\pm 1.0\%$ (y-pol.). These results confirm that the proposed logic gate can simultaneously achieve amplitude weighting and phase-dependent routing, thereby establishing the basis for cascaded in-fibre linear logic circuits.

Experimental verification of MMI-assisted coupler

We fabricated the device by femtosecond-laser inscription in a two-core fibre (core radius $4.5 \mu\text{m}$, spacing $40 \mu\text{m}$), writing an MMI region in the cladding (Fig. 4). The fabricated MMI region targeted C-band 50:50 splitting with dimensions $24 \mu\text{m}$ (W), $6200 \mu\text{m}$ (L), and $9 \mu\text{m}$ (H). A shorter device was used than in the previous design to reduce fabrication time and excess loss in the laser-inscribed waveguide. Figure 5 compares the measured and calculated splitting ratios across the C band. At 1550 nm, the measured splitting is $50 \pm 5\%$ with an excess loss of 1.19 dB. Toward shorter wavelengths, the measured splitting departs from the calculation, which is attributed to shifts in the phase-matching conditions due to effective-index dispersion and fabrication tolerances. This finding indicates the need for further geometry/index optimisation to ensure a flatter spectral response.

Finally, phase-dependent output-core switching was verified. The inputs to the two

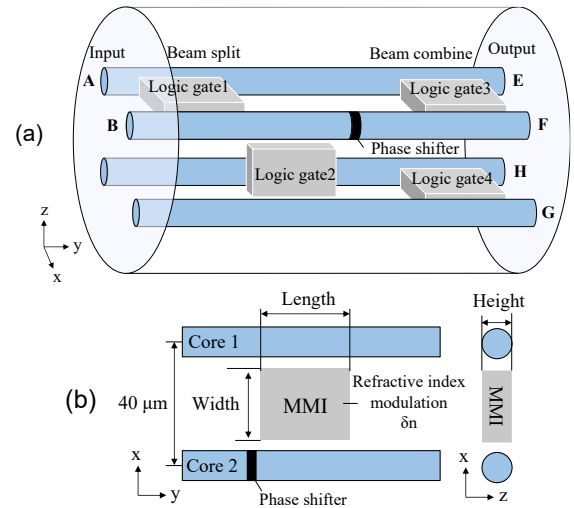


Fig. 1: Proposed in-fibre logic circuit. (a) Circuit layout with inputs A and B, MMI1–MMI4, phase shifter, and outputs E–H. (b) Proposed MMI-assisted coupler and design parameters (W , L , H , and δn).

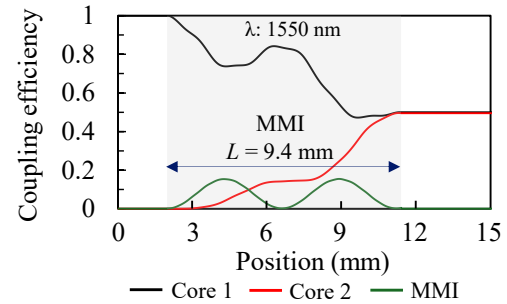


Fig. 2: BPM results of the MMI-assisted coupler with near-50:50 1×2 splitting. For $W = 22 \mu\text{m}$ and $L = 9400 \mu\text{m}$, the loss is 0.026 dB and the splitting-ratio deviation from 50:50 is within $\pm 2.4\%$ (x-pol.) and $\pm 1.0\%$ (y-pol.).

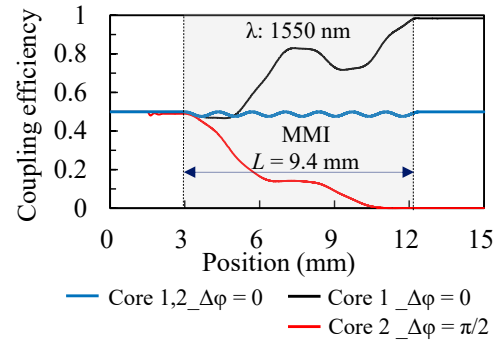


Fig. 3: BPM results of the MMI-assisted coupler with phase-dependent 2×2 output-core switching. The output is nearly balanced at $\Delta\phi = 0$ and concentrated in Core 1 at $\Delta\phi = \pi/2$.

cores were generated using a 3-dB coupler, and $\Delta\phi$ was adjusted by fibre bending. In a practical implementation, this role can also be performed by an in-fibre phase shifter. Figure 6 shows the output from the two cores as the input phase difference was varied. At 1550 nm, the outputs were nearly balanced at $\Delta\phi \approx 0$, whereas the output core switched when $\Delta\phi$ was changed to $\pi/2$ and π . These results showed good agreement with the BPM calculations, confirming

that the MMI-assisted coupler can be utilized as a linear logic gate.

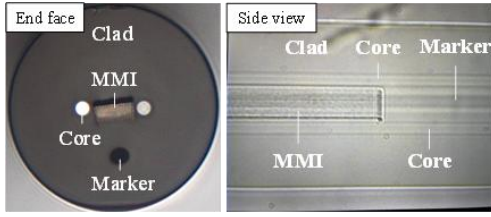


Fig. 4: Fabrication of MMI-assisted coupler in two-core fibre showing end-face and side views.

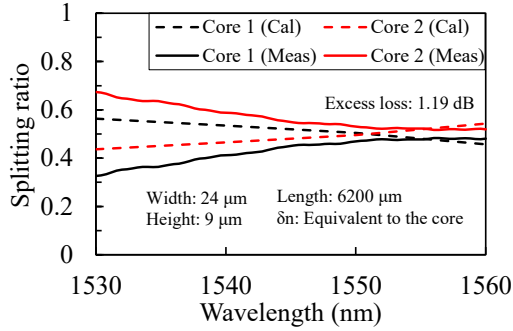


Fig. 5: Measured and calculated splitting ratio of the MMI-assisted coupler in the C band.

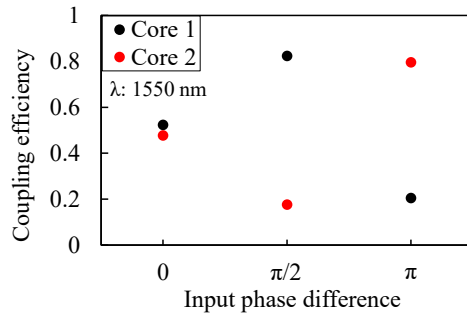


Fig. 6: Verification of phase-dependent switching by adjusting the relative input phase.

Applicability to 2×4 in-fibre linear logic circuit

As an application of the proposed 2×2 logic gate, we investigate the 2×4 in-fibre linear logic circuit shown in Fig. 1(a). To achieve a clear extinction ratio for each output, this circuit is implemented within a 4-core optical fibre, integrating four 2x2 logic gates and a single in-fibre phase control section.

In this configuration, logic gate 1 provides asymmetric amplitude weighting of the two input signals, defining the relative contributions of inputs A and B. Logic gates 2–4 operate as balanced interferometric combiners, where the weighted signal paths interfere with a controlled phase. Consequently, different interference conditions are established at each output core, and as summarized in Table 1, the four outputs E to H exhibit logical characteristics corresponding to OR, XOR, and AND.

The central phase shifter sets the relative phase between the two signal paths, enabling

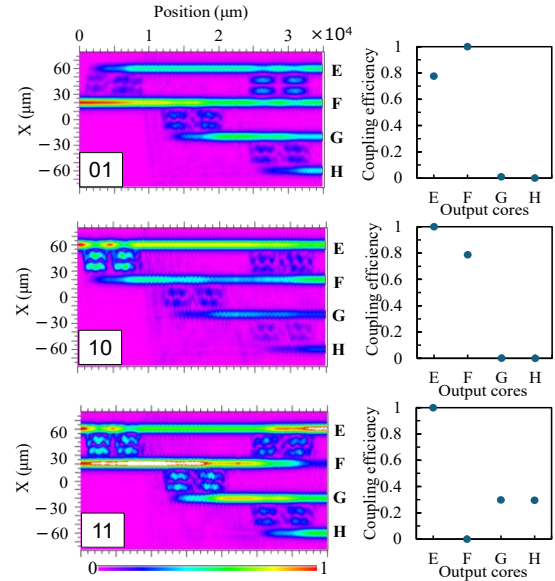


Fig. 7: Simulated output states of proposed in-fibre logic circuit.

Tab. 1: Truth table classified from calculation results.

Input (AB)	Output (EFGH)			
0 0	0	0	0	0
0 1	1	1	0	0
1 0	1	1	0	0
1 1	1	0	1	1
Output cores	E F G H (OR)(XOR)(AND)(AND)			

selective constructive and destructive interference at downstream couplers. In the model, it is represented by a refractive index modulated region with $\delta n = 0.002$, a radius of $4.5 \mu\text{m}$, and a length of $640 \mu\text{m}$, corresponding to a phase shift of 246° at 1550 nm .

Figure 7 shows the simulated normalized output intensities at the four output cores (E–H). To maximize the extinction ratio, the splitting ratio of MMI1 is set to 70:30, while those of the other couplers are set to 50:50. By defining a normalized intensity of 0.2 or higher as logic “1,” the input–output relationships in Table 1 are obtained. These results demonstrate that a linear optical logic circuit can be constructed within a fibre using the MMI-assisted coupler as a basic building block.

Conclusions

We demonstrated a femtosecond laser inscribed MMI-assisted coupler in a two-core fibre as a 2×2 building block for in-fibre linear logic. At 1550 nm , the device shows $\sim 50:50$ splitting ($50 \pm 5\%$) with 1.19 dB excess loss and $\Delta\phi$ -controlled output-core switching, consistent with numerical predictions. Building on these functions, a cascaded 2×4 architecture is shown to generate parallel OR/XOR/AND outputs selectable by output core, supporting MCF-integrated linear processing.

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